

Programming Paradigm Hierarchy — Classify Paradigms — Answers

Q1. Categorize Java, Prolog, SQL, and Python based on paradigm.

- Java: primarily object-oriented, also supports procedural (static methods) and functional (lambdas, Stream API) styles -- a multi-paradigm imperative/OOP language.
- Prolog: a logic programming language, declarative in nature; programs are written as facts and rules, and computation happens through logical inference.
- SQL: a declarative language, specifically used for querying and manipulating relational databases; the programmer states what data is needed, not how to retrieve it.
- Python: a multi-paradigm language supporting procedural, object-oriented, and functional styles, similar in spirit to Java but more dynamically typed and flexible.

In summary, Java and Python are both multi-paradigm languages centered on imperative and object-oriented programming, while Prolog is a logic language and SQL is a purely declarative, data-query language.

Q2. Create a table showing imperative, declarative, logic, and functional features.

The table below summarizes the defining features of each paradigm category.

Paradigm	Focus	Example Feature	Example Language
Imperative	How to do it (steps)	Loops, assignments	Java, C
Declarative	What is needed	Queries, constraints	SQL
Logic	Facts and rules	Inference, unification	Prolog
Functional	Function evaluation	Pure functions, lambdas	Java (lambdas), Haskell

Imperative programming describes explicit step-by-step instructions that change program state. Declarative programming describes the desired result, leaving the how to the underlying system. Logic programming is a specialized declarative approach based on facts, rules, and inference. Functional programming treats computation as the evaluation of pure functions without changing shared state.

Q3. Explain the hierarchical relationship between paradigms.

Programming paradigms can be organized in a broad hierarchy, starting from the most general division and narrowing into specific styles:

- At the top level, paradigms split into imperative (how to do something, using explicit state and control flow) and declarative (what result is wanted, without specifying exact steps).

- Under the imperative branch: procedural programming (plain sequences of instructions and procedures) and object-oriented programming (instructions organized around objects) are both imperative sub-paradigms, since both rely on changing program state step by step.
- Under the declarative branch: functional programming (computation as function evaluation) and logic programming (computation as logical inference over facts and rules) are both declarative sub-paradigms.
- Database query languages such as SQL sit within the declarative branch, as a specialized form focused on data retrieval and manipulation.

Java sits mainly within the imperative branch (procedural and object-oriented), but also borrows functional-style features (lambdas, streams) from the declarative branch, which is why it is described as multi-paradigm rather than belonging to a single node in this hierarchy.